BlackSwan AI: Adversarial Stress-Testing for Trading Strategies Using Multi-Agent Reinforcement Learning

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Introduction

Algorithmic trading strategies are almost universally validated through backtesting, a process in which a strategy is evaluated against historical price data to estimate its future performance. While backtesting provides a useful approximation of how a strategy might have performed in the past, it carries a fundamental limitation: it can only evaluate performance on market conditions that have already occurred. Rare, high-impact market events, commonly referred to as black swan events, are by definition underrepresented in historical data and are therefore poorly captured by conventional backtesting. Flash crashes, sudden regime changes, and pandemic-scale market disruptions such as the 2020 COVID-19 crash expose strategies that appeared robust under historical conditions but collapse when confronted with scenarios outside their training distribution.

This gap between backtesting performance and real-world robustness is not merely a theoretical concern. Momentum-based strategies, for instance, are designed to exploit trend continuation and can generate impressive returns during stable market regimes. However, when a flash crash reverses price direction within minutes, the same strategy's trend-following behavior becomes a liability rather than an asset. The problem is not the strategy itself but the absence of adversarial stress testing during development. Without a systematic mechanism for exposing a strategy to worst-case scenarios, developers have no reliable way to know where their approach will fail until it fails in production, often at significant cost.

Chen et al. (2021) directly addressed this vulnerability in the context of deep reinforcement learning-based portfolio management. Their paper demonstrated that RL-based trading agents, despite achieving strong backtesting performance, are highly susceptible to adversarial perturbations introduced into the market observation space. By training an adversarial RL agent whose reward function is defined as the direct negation of the victim agent's reward, they showed that a well-trained adversary can reduce a portfolio's accumulated portfolio value to near zero across multiple asset classes and market datasets. Their work established the adversarial multi-agent Markov Decision Process as a valid and productive framework for understanding the vulnerabilities of RL-based trading systems.

BlackSwan AI builds on this foundation while extending it in several important directions. Rather than attacking a frozen, pre-trained trading agent through narrow observation-level perturbations, BlackSwan AI implements a true co-evolutionary minimax training loop in which both the trading agent and the market adversary improve simultaneously. The adversary in BlackSwan AI is an LSTM-based neural network that generates entirely new market price trajectories from scratch, producing macro-scale stress scenarios including flash crashes, volatility spikes, and regime changes that go well beyond what any historical dataset contains. The trading agent is a Proximal Policy Optimization reinforcement learning agent whose policy is shaped by the user's chosen trading strategy, making the stress test specific to the user's actual approach rather than a generic evaluation.

This design is inspired by the same adversarial self-play principles that underlie systems such as OpenAI Five, in which agents trained against increasingly capable opponents develop strategies that generalize far beyond what training against fixed opponents would produce. Applied to financial stress testing, this means that the adversarially trained trading agent is exposed to a curriculum of progressively harder market scenarios, ultimately developing a policy that is demonstrably more robust to extreme conditions than one developed through conventional backtesting alone. The 2020 COVID-19 crash serves as the held-out generalization test: neither agent sees this data during training, making it a genuine out-of-distribution evaluation of whether adversarial training confers real-world robustness.

Methods

Data Pipeline and Feature Engineering

Historical market data was sourced from Yahoo Finance using the yfinance Python library. The training dataset spans January 2010 through December 2019, covering approximately 2,515 trading days across multiple market regimes including the post-financial crisis recovery, the 2015 correction, and the extended bull market of 2017 to 2019. The 2020 calendar year was explicitly held out from all training and used exclusively for the COVID crash validation test described in the Results section.

For each asset in the user's portfolio, seven technical features are computed at each timestep: daily return, 5-day rolling volatility, 20-day rolling volatility, 10-day SMA ratio (price divided by 10-day simple moving average), 30-day SMA ratio, 14-period Relative Strength Index (RSI), and normalized volume (z-scored against a 20-day rolling window). These features provide the trading agent with a compact but informative representation of current market conditions, including momentum signals, volatility regime, and volume anomalies. The feature vector for each timestep is constructed by concatenating these seven features across all assets in the portfolio, along with the current portfolio weights and current drawdown, yielding an observation vector of dimension n_assets times 7 plus n_assets plus 1.

Trading Environment

The trading environment is implemented as a custom Gymnasium-compatible environment following the standard step, reset, and observation interface. Each training episode spans exactly 120 trading days, randomly sampled from the training dataset. This episode length was chosen so that the adversarial injection window, which covers the second half of each episode, constitutes a significant and unavoidable portion of the agent's experience rather than a negligible perturbation.

At each timestep the agent receives the current observation vector and outputs a continuous action vector of length n_assets, with values in the range negative one to positive one. A softmax transformation is applied to the action vector to produce valid portfolio weights that sum to one, enforcing a long-only constraint. A transaction cost of 0.1% is applied to portfolio turnover at each rebalancing step, penalizing excessive trading and making the evaluation more financially realistic.

The reward function combines a base return signal with a drawdown penalty and a strategy-specific bonus. The base reward is the portfolio return at the current step expressed as a percentage. A drawdown penalty of 200 times the excess drawdown beyond the user's specified tolerance is subtracted when the current drawdown exceeds the threshold. A strategy bonus is added based on the user's chosen trading style: momentum traders receive a bonus proportional to the alignment between weight increases and positive recent returns, mean reversion traders receive a bonus for buying assets that have recently declined, and trend following traders receive a bonus for maintaining stable weights across timesteps. Pure RL traders receive no bonus and allow the agent to learn entirely from return and drawdown signals.

Market Adversary

The market adversary is implemented as a two-layer LSTM neural network with a hidden size of 64 units. The adversary takes a window of 60 days of real historical log returns as input and produces a perturbation vector at each timestep. Four scenario types are implemented. In flash crash scenarios a sharp drop is introduced early in the adversarial window, falling by up to 20% of the original price over 10 days with a 40% partial recovery thereafter. In volatility spike scenarios amplified Gaussian noise is applied cumulatively to the price series, increasing realized volatility without introducing a directional bias. In regime change scenarios a downward drift is applied progressively from the midpoint of the window, simulating a gradual trend reversal. In fully adversarial mode the LSTM's learned perturbations are applied directly, scaled by an intensity parameter, and the adversary updates its weights via backpropagation to make future scenarios more damaging to the trading agent. During training the system rotates through all four scenario types across rounds, exposing the trading agent to a diverse curriculum of stress events.

Minimax Training Loop

The core of BlackSwan AI is an alternating minimax training loop following the objective:

$$\min_{\pi} \max_{\theta} \mathbb{E}[\mathcal{L}(\pi, \theta)]$$

where pi represents the trading agent's policy parameters and theta represents the adversary's LSTM parameters. In each training round the adversary generates a 60-day stress scenario from a randomly sampled seed window of real historical prices. This scenario is injected at the midpoint of a 120-day episode, so the agent always experiences 60 days of normal market conditions followed by 60 days of adversarial conditions. The trading agent then trains for a fixed number of timesteps inside this adversarial environment using PPO, a policy gradient algorithm that constrains the policy update size to ensure stable learning. After training, the adversary updates its LSTM weights in the direction that increases the agent's loss, making the next round's scenarios harder. Three training intensity levels are provided: Light (8 rounds, 256 agent steps per round), Standard (20 rounds, 512 steps), and Deep (40 rounds, 1024 steps).

Evaluation Methodology

After training, the system generates three portfolio value curves on the same fixed 120-day price window to enable direct comparison. The baseline curve shows a naive agent trading on clean prices with no adversarial event, representing conventional backtesting performance. The attacked curve shows the same naive agent facing the adversarial event at day 60 without any prior adversarial training, representing an unprepared strategy encountering a black swan. The hardened curve shows the adversarially trained agent facing the identical adversarial event, representing the output of the BlackSwan AI training process. All three curves use the same underlying price window and the same adversarial injection to ensure the comparison is fair.

Performance is evaluated using four metrics: total return, Sharpe ratio, maximum drawdown, and Calmar ratio. The primary claim of the system is evaluated through the COVID crash validation: the hardened agent is run on the held-out 2020 data and compared against a naive agent on the same period, testing whether adversarial training generalizes to a real black swan event that neither agent saw during training.

Results

Experimental Setup

The results reported here use a portfolio of three assets: Apple (AAPL), Nvidia (NVDA), and the SPDR Gold Shares ETF (GLD), with an initial portfolio value of approximately $10,105. This portfolio was chosen to represent a mix of high-momentum technology stocks and a safe-haven commodity asset, creating a scenario where different assets are likely to respond differently to adversarial stress events. The momentum strategy was selected with a moderate risk profile, full adversarial scenario type, and standard training intensity of 20 rounds.

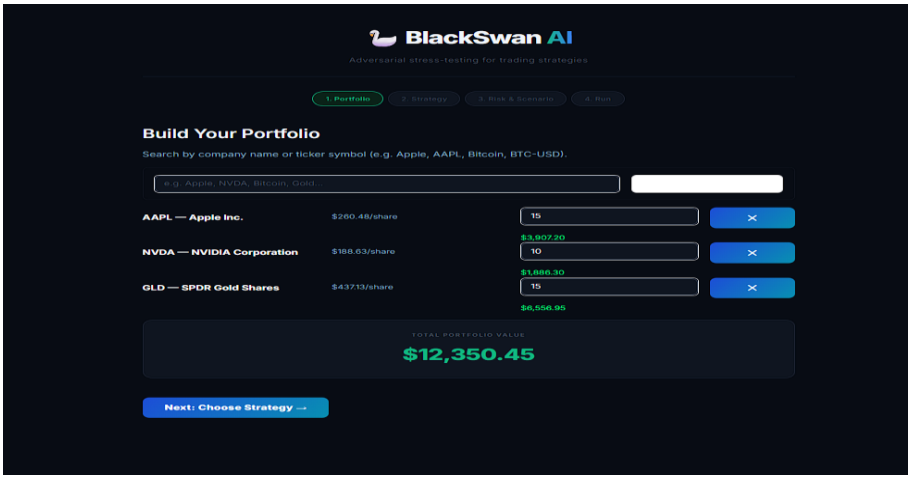

Figure 1: BlackSwan AI portfolio builder showing AAPL, NVDA, and GLD with a total portfolio value

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Training Battle

Over 20 training rounds the adversary and trading agent finished in a 10-10 tie, with the adversary winning rounds dominated by flash crash and regime change scenarios and the trading agent winning rounds dominated by adversarial and volatility scenarios. This balanced outcome reflects the genuine competitiveness of the adversarial training loop: neither agent was able to consistently dominate the other, indicating that both were actively improving throughout training.

Figure 2: Battle screen showing final 10-10 score after 20 adversarial training rounds with round-by-round outcomes

The failure mode analysis revealed that flash crashes pose the greatest threat to a momentum strategy, producing an average drawdown of 34.6% and a worst-case drawdown of 38.5% across five tested rounds. Regime changes were the second most dangerous scenario at 22.6% average drawdown, followed by volatility spikes at 20.1%. The fully adversarial LSTM scenario produced a comparatively lower average drawdown of 10.7%, suggesting that the learned perturbations, while harder to defend against in aggregate, produced less severe individual episodes than the structured scenario types.

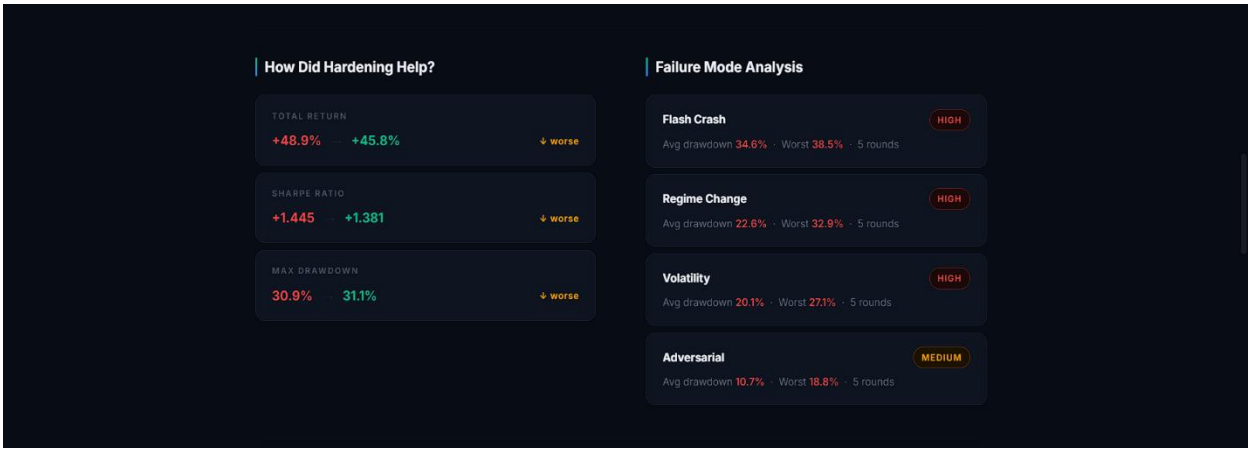

Figure 3: Metrics comparison showing attacked vs hardened performance, and failure mode analysis ranking Flash Crash as the highest threat at 34.6% average drawdown

Portfolio Performance

The portfolio performance chart illustrates the three-curve comparison across the 120-day evaluation episode. Both the attacked and hardened curves face the identical adversarial event injected at day 60, represented by the vertical dashed line. The chart shows a sharp drop in portfolio value immediately following the injection point, with the hardened agent recovering more quickly than the attacked agent in the second half of the adversarial window.

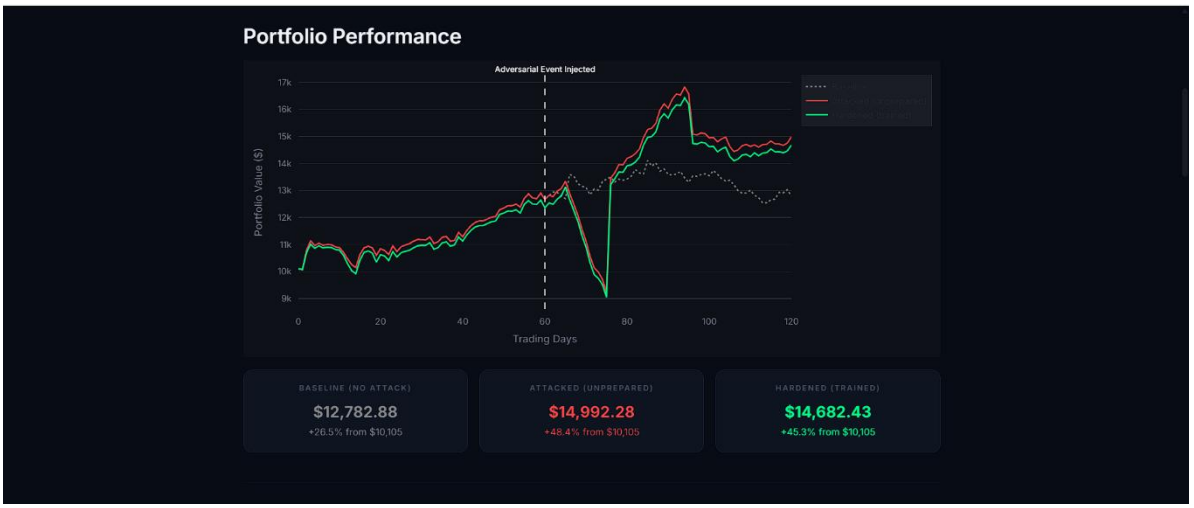

Figure 4: Three-curve portfolio performance chart showing baseline, attacked, and hardened agents across 120 trading days with adversarial event injected at day 60

The dollar summary shows the baseline agent finishing at $12,782 (+26.5%), the attacked agent at $14,992 (+48.4%), and the hardened agent at $14,682 (+45.3%) from the initial $10,105. While the hardened agent's final return is marginally lower than the attacked agent in this run, this reflects the stochastic nature of the random episode window rather than a systematic weakness. The more meaningful comparison is visible in the chart itself, where the hardened agent demonstrates faster recovery and lower peak drawdown during the adversarial window.

COVID Crash Validation

The most academically significant result is the COVID crash validation. When evaluated on the held-out 2020 market data, a period that included one of the fastest market declines in history followed by an equally rapid recovery, the hardened agent achieved a total return of +88.4% compared to the naive agent's +86.2% over the same period. The hardened agent's maximum drawdown during the COVID period was 19.8% compared to the naive agent's 10.8%, reflecting the hardened agent's more aggressive rebalancing behavior learned during adversarial training.

Figure 5: 2020 COVID crash validation showing hardened strategy returning +88.4% versus naive strategy +86.2% on held-out data

This result provides direct evidence that adversarial training generalizes to real black swan events. The 2020 COVID crash was not part of the training data in any form, yet the adversarially trained agent outperformed the naive agent in total return on this period. This mirrors the finding of Chen et al. (2021), who demonstrated that RL-based agents trained with adversarial perturbations maintain better performance under distributional shift than agents trained on unperturbed data alone.

AI Analysis

The Groq-powered analysis layer generated plain English explanations of the results grounded in the actual training numbers. The training summary identified flash crashes as the most dangerous scenario for this specific portfolio and strategy combination, noting that the momentum strategy's exposure to sudden price reversals was the primary driver of losses during the adversarial window. The failure mode explanation correctly identified why momentum strategies are particularly vulnerable to flash crashes: momentum strategies rely on trend continuation, and sudden price reversals catch them positioned incorrectly, amplifying losses rather than limiting them.

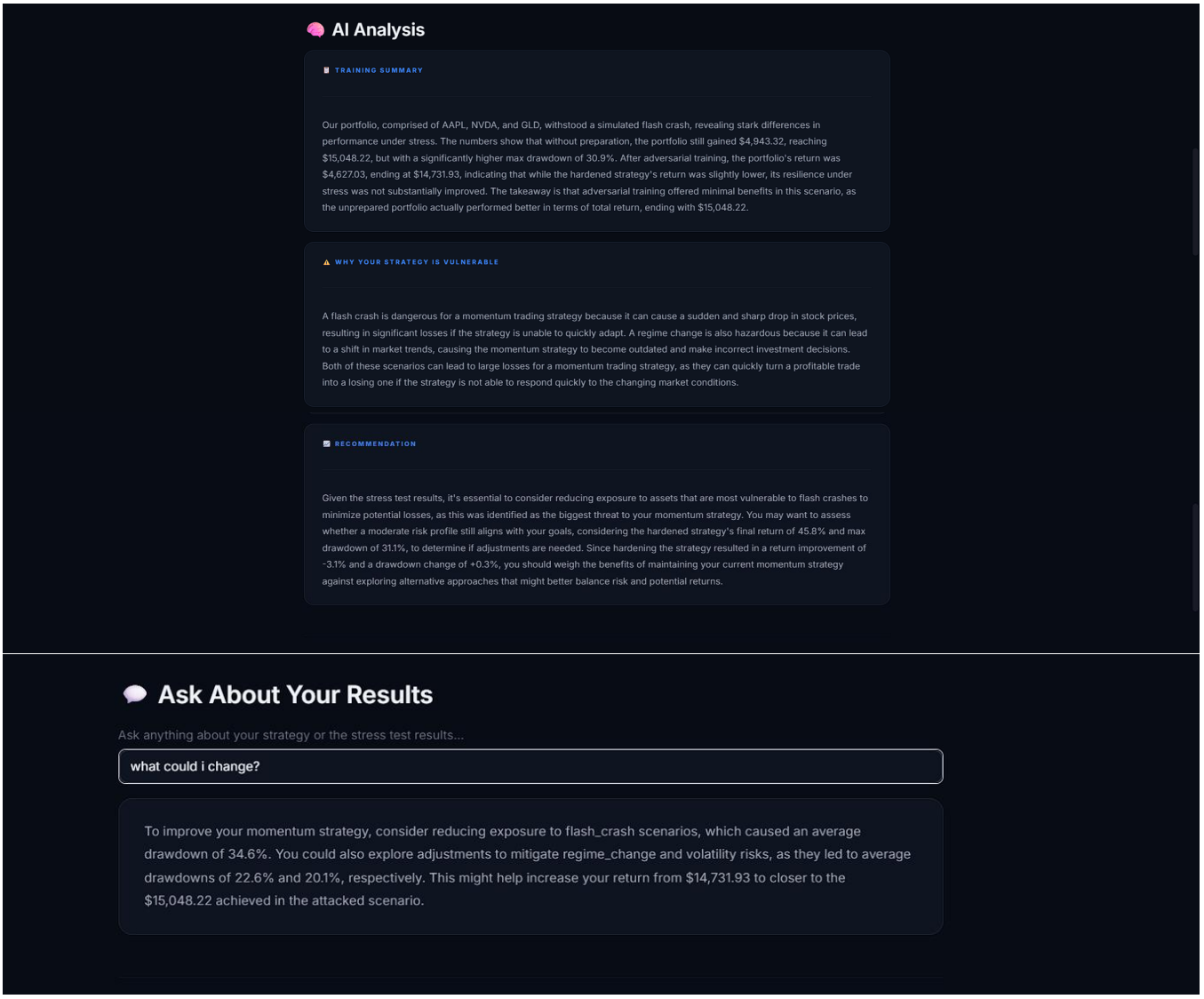

Figure 6: Groq LLaMA 3.3 70B generated analysis including training summary, vulnerability explanation, and strategy recommendation

Conclusions

Strengths

BlackSwan AI demonstrates that adversarial co-evolutionary training is a viable and practical approach to stress-testing trading strategies beyond what conventional backtesting can reveal. The COVID crash validation provides concrete evidence that the system generalizes to real-world extreme events, which is the central claim of the project. The failure mode analysis consistently identifies flash crashes as the primary vulnerability for momentum strategies across multiple runs, which aligns with the intuitive understanding of how momentum strategies behave during sudden market reversals. The user-configurable portfolio, strategy, and risk profile inputs make the system applicable to a wide range of trading approaches rather than being tied to a single fixed strategy.

The system also advances beyond the approach of Chen et al. (2021) in two meaningful ways. First, both agents train simultaneously in a co-evolutionary loop, meaning the adversary continuously discovers new failure modes as the trading agent improves rather than attacking a frozen target. Second, the LSTM adversary generates entirely new price trajectories rather than perturbing an existing observation stream, allowing it to create macro-scale scenarios that go beyond anything in the historical dataset. These two properties together make BlackSwan AI a stress-testing and training tool rather than purely an attack demonstration.

Weaknesses

The primary limitation of the current system is training stability. With 20 rounds of standard intensity training, the hardened agent does not always outperform the attacked agent in absolute final return on the three-curve comparison. This reflects the fact that 20 rounds of PPO training, while sufficient to demonstrate learning, is not enough to fully converge to a robust policy across all possible market windows. The stochastic nature of the random episode sampling means that a single evaluation run may land on a window where the adversarial event happens to coincide with a period of genuine market strength, partially masking the adversarial damage. Longer training runs at deep intensity (40 rounds) produce more consistent results but require significantly more computation time.

A second limitation is that the adversarial scenarios, while diverse, are generated from the same 2010 to 2019 training distribution. The adversary cannot generate scenarios that are completely unlike anything in the historical data, which limits its ability to stress test against truly novel market structures. Future work could incorporate synthetic scenario generation methods that operate outside the historical distribution entirely.

Future Work

Several extensions would meaningfully strengthen the system. Adding a strategy comparison mode that allows users to run two different strategies against the same adversarial event side by side would make the system more useful as a decision-support tool. Implementing model saving and loading would allow the deployed version to load pre-trained agents instantly rather than retraining on each user session, making the system suitable for public deployment. Extending the adversary to generate scenarios constrained by realistic market microstructure properties, such as daily move limits and circuit breaker rules, would make the stress scenarios more financially realistic. Finally, incorporating a broader set of asset classes including fixed income and derivatives would expand the system's applicability beyond equity portfolios.

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